GMAT Quantitative Reasoning

Word Problems

Hard / Very Hard Problem Modeling

Challenging exercises focused on modeling complex word problems. These problems require careful translation of verbal descriptions into mathematical equations, multi-step reasoning, and systematic solution approaches. Work through the questions first, then check the detailed solutions.

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EDUCATIONAL PURPOSE: This material is intended solely for educational purposes to help students prepare for standardized tests.

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1. Part I: Exercises

Work through each problem carefully. Focus on setting up the problem correctly before attempting to solve it. These problems are designed to challenge your problem-modeling skills at a high level.

Difficulty Level: Hard to Very Hard

These problems require:

- Careful translation of words into mathematical expressions
- Multiple variables and equations working together
- Strategic organization of complex information
- Systematic case analysis and constraint handling

1.1. Section A: Advanced Rate Problems

1.1.1. Exercise A1: Multi-Leg Journey with Variable Speeds

A car travels from City A to City B. For the first third of the distance, it travels at 60 km/h. For the second third, it travels at 80 km/h. For the final third, it travels at 120 km/h.

Questions:

1. Set up expressions for the time taken for each segment in terms of the total distance D.
2. Calculate the total time for the journey in terms of D.
3. Find the average speed for the entire journey. Explain why this differs from the arithmetic mean of the three speeds.
4. If the total distance is 360 km and the car had instead traveled at a constant speed equal to the arithmetic mean of the three speeds, would it have arrived earlier or later? By how much?

1.1.2. Exercise A2: Pursuit and Meeting Problems

Two trains leave stations A and B simultaneously, heading toward each other. Train 1 travels at 80 km/h and Train 2 at 100 km/h. The distance between stations is 450 km.

Questions:

1. Model the position of each train as a function of time t hours after departure.
2. At what time and at what location do the trains meet?
3. A bird starts at Train 1 and flies back and forth between the trains at 150 km/h until the trains meet. Without calculating each leg of the bird's journey, find the total distance the bird travels.
4. If Train 1 departs 30 minutes before Train 2, at what position do they meet?

1.1.3. Exercise A3: Round-Trip Average Speed

A cyclist rides uphill at 12 km/h and returns downhill on the same path at 36 km/h.

Questions:

1. Prove that the average speed for the round trip is 18 km/h, regardless of the distance. Derive this using the harmonic mean formula.
2. Derive a general formula for the average round-trip speed given uphill speed u and downhill speed d: show that the result is $\frac{2ud}{u+d}$.

3. The cyclist can alternatively take a flat route at 20 km/h. If the hilly route is 25% shorter (one way), which route is faster for a round trip?
4. Generalize: for what percentage difference in route length would the two options take exactly the same time?

1.1.4. Exercise A4: Stream and Current Problems

A boat can travel 30 km downstream in the same time it takes to travel 18 km upstream.

Questions:

1. Let b be the boat's speed in still water and c be the current speed. Set up an equation relating these quantities to the given information.
2. Solve for the ratio $b : c$.
3. If the boat takes 2 hours to travel 48 km downstream, find the exact values of b and c.
4. How long would it take the boat to travel 80 km upstream and return downstream to its starting point?

1.1.5. Exercise A5: Circular Track Problems

Two runners start at the same point on a 400-meter circular track. Runner A runs clockwise at 5 m/s, and Runner B runs counterclockwise at 3 m/s.

Questions:

1. How long until they first meet again at the starting point? (LCM approach)
2. How long until they first meet anywhere on the track? (Relative speed approach)
3. How many times do they meet (excluding the start) in the first 10 minutes?
4. If Runner A stops after 4 minutes, at what position on the track is Runner A, and how long after Runner A stops until Runner B reaches that position?

1.2. Section B: Advanced Work Problems

1.2.1. Exercise B1: Multiple Workers with Variable Rates

Worker A can complete a job in 10 days. Worker B can complete it in 15 days. Worker C can complete it in 30 days.

Questions:

1. Express each worker's daily rate as a fraction of the job.
2. If all three work together, how many days does the job take? Set up and solve the equation.
3. If A and B work together for 3 days, then B and C complete the remaining work, how many total days does the job take?
4. Is it possible to design a schedule where each worker works exactly 5 days and the job is completed exactly? Prove your answer algebraically.

1.2.2. Exercise B2: Work with Efficiency Changes

Machine A produces 120 widgets per hour. Machine B produces 80 widgets per hour. However, Machine A requires 15 minutes of maintenance every hour, while Machine B operates continuously.

Questions:

1. Calculate the effective hourly production rate for each machine.
2. If both machines operate for a standard 8-hour shift, how many total widgets are produced?
3. After 4 hours of operation, Machine B's efficiency drops by 25%. How many widgets does Machine B produce in the full 8-hour shift?
4. A rush order requires 1,400 widgets. If both machines start simultaneously (with Machine B's efficiency drop after 4 hours still applying), how long until the order is complete?

1.2.3. Exercise B3: Pipes and Tanks

A tank has two inlet pipes and one outlet drain. Pipe A can fill the empty tank in 4 hours. Pipe B can fill it in 6 hours. The drain can empty a full tank in 8 hours.

Questions:

1. Express each pipe's rate (positive for filling, negative for draining) as a fraction of the tank per hour.
2. If all three are open simultaneously, how long does it take to fill an empty tank?
3. The tank is initially $\frac{1}{3}$ full. Pipe A and the drain are both opened. After 2 hours, Pipe B is also opened. How long from the start until the tank is completely full?
4. Starting with an empty tank and all three open, after how many hours should the drain be closed so that the tank fills at exactly $t = 6$ hours?

1.2.4. Exercise B4: Work Rate Ratios

Two workers, X and Y, work on a project. X works at three times the rate of Y.

Questions:

1. If they complete the job together in 6 days, find each worker's individual completion time.
2. After 2 days of working together, X leaves. How many additional days does Y need to finish alone?
3. Starting over: X works alone for the first k days, then Y joins and they finish together. If the total time is 10 days, find k.
4. Generalize: if X works at n times the rate of Y, and together they complete the job in T days, express each worker's individual completion time in terms of n and T.

1.3. Section C: Mixture Problems

1.3.1. Exercise C1: Multi-Component Solutions

A chemist has three solutions: Solution A with 15% acid, Solution B with 40% acid, and Solution C with 75% acid.

Questions:

1. To create 100 mL of a 35% acid solution using only A and B, how many mL of each are needed?
2. Is it possible to create a 35% acid solution using equal volumes of all three solutions? Calculate the resulting concentration.
3. The chemist needs 300 mL of exactly 50% acid using A, B, and C. If the volume of C used equals the volume of A used, how much of each solution is needed?
4. What is the complete range of acid concentrations achievable using mixtures of A and C only? Prove your answer.

1.3.2. Exercise C2: Repeated Replacement

A 200-liter tank contains a 60% alcohol solution.

Questions:

1. If 40 liters are removed and replaced with pure water, what is the new alcohol concentration?
2. If this process (remove 40L, replace with water) is repeated a second time, what is the concentration?
3. Derive a formula for the concentration after n such operations.
4. How many complete operations are needed to reduce the concentration to below 20%?

1.3.3. Exercise C3: Alloy and Mixture Value Problems

A jeweler has two gold alloys: Alloy X is 50% gold, and Alloy Y is 80% gold.

Questions:

1. How much of each alloy is needed to create 500 grams of a 70% gold alloy?
2. If the jeweler has only 150 grams of Alloy Y available, what is the maximum amount of 70% gold alloy that can be created?
3. The jeweler melts some 70% gold alloy with pure gold. How much pure gold must be added to 400 grams of 70% gold alloy to create an 85% gold alloy?
4. Prove that any concentration strictly between 50% and 100% can be achieved using Alloy X and pure gold, but concentrations at or below 50% cannot be achieved by adding pure gold to Alloy X.

1.3.4. Exercise C4: Pricing and Mixture Economics

A merchant mixes three types of coffee: Type A at $8/kg, Type B at $14/kg, and Type C at $24/kg.

Questions:

1. If 4 kg of A, 5 kg of B, and 1 kg of C are mixed, what is the cost per kg of the mixture?
2. The merchant wants to create a 15 kg mixture worth $16/kg using only A and C. How much of each type is needed?
3. What is the range of possible prices per kg for any mixture using only A and B?
4. The merchant has unlimited A and C, but only 3 kg of B. For a 12 kg mixture, what is the range of achievable prices per kg?

1.4. Section D: Age Problems

1.4.1. Exercise D1: Multiple Time References

Currently, Maria is four times as old as her daughter Elena. In 16 years, Maria will be twice as old as Elena.

Questions:

1. Set up a system of equations using Maria's current age M and Elena's current age E.
2. Solve for their current ages.
3. How many years ago was Maria seven times as old as Elena?
4. In how many years will the sum of their ages first exceed 100?

1.4.2. Exercise D2: Age Ratios

Three siblings have ages that are currently in the ratio 1 : 2 : 4. The sum of the squares of their ages is 336.

Questions:

1. Express each sibling's age in terms of a common variable k.
2. Set up and solve an equation to find each sibling's current age.
3. In how many years will their ages be in the ratio 2 : 3 : 5?
4. Is there any future time when their ages will be in the ratio 3 : 4 : 5? Prove or disprove algebraically.

1.4.3. Exercise D3: Past and Future Conditions

Five years ago, a father was 7 times as old as his son. Five years from now, the father will be 3 times as old as his son.

Questions:

1. Let F and S be the current ages. Set up two equations from the given conditions.
2. Solve for the current ages of the father and son.
3. At what age was the father exactly 5 times as old as his son?
4. Prove that there is no age at which the father was exactly 10 times as old as his son (assuming both have been alive since birth).

1.4.4. Exercise D4: Combined Age Conditions

The sum of the ages of a grandfather, father, and son is 120 years. The grandfather is twice as old as the father. Ten years ago, the grandfather was three times as old as the father was then.

Questions:

1. Let G, F, and S be the current ages. Set up three equations.
2. Solve for all three ages.
3. What was the son's age when the father was the same age as the son is now?
4. In how many years will the sum of the father's and son's ages equal the grandfather's age at that time?

1.5. Section E: Complex Multi-Step Problems

1.5.1. Exercise E1: Distance with Multiple Travelers

Alice and Bob start from the same point and walk in the same direction. Alice walks at 4 km/h. Bob starts 30 minutes later and walks at 6 km/h.

Questions:

1. At what time after Alice starts does Bob catch up to her?
2. How far from the starting point does this occur?
3. If Charlie starts at the same time as Alice but walks in the opposite direction at 5 km/h, how far apart are Bob and Charlie when Bob catches Alice?
4. At what time after Alice starts are Alice and Charlie exactly 20 km apart?

1.5.2. Exercise E2: Work with Changing Team Composition

A project requires a certain amount of work. Person A working alone can complete it in 20 days. Person B working alone can complete it in 30 days.

Questions:

1. A starts the project alone. After 8 days, A is joined by B. How many total days does the project take?
2. In the same scenario, what fraction of the work does each person complete?
3. Redesign: A and B start together, but after some number of days d, A leaves and B finishes alone. If the total project time is 15 days, find d.
4. What value of d (where A works alone first, then B joins) minimizes the total project time while ensuring each person does at least 25% of the work?

1.5.3. Exercise E3: Mixture with Constraints

A container initially holds 100 liters of a 30% salt solution. A second container holds a 70% salt solution.

Questions:

1. How much solution must be transferred from the second container to the first to create a 45% salt solution in the first container?
2. After this transfer, what is the total volume in the first container?
3. Instead, suppose we want the first container to have exactly 50% salt concentration but must maintain exactly 100 liters total volume. How much solution must be removed from the first container and replaced with solution from the second?
4. Generalize: starting with V liters of c_1% solution, derive a formula for the amount of c_2% solution (where $c_2 > c_1$) needed to achieve target concentration c_t%.

1.5.4. Exercise E4: Rate Problem with Delays

A train travels from Station A to Station B, a distance of 300 km. Due to a signal problem, the train stops for 20 minutes after traveling 100 km, then resumes at its original speed.

Questions:

1. If the train's speed is 80 km/h, what is the total travel time from A to B?
2. The train was supposed to arrive at B exactly 4 hours after leaving A. What constant speed would have been required (without the delay) to arrive on time?
3. With the 20-minute delay after 100 km, what speed would the train need on the remaining 200 km segment to still arrive at the originally scheduled time (4 hours after departure)?
4. If the train increases its speed by 25% for the final 200 km segment (after the delay), what original speed v results in a total travel time of exactly 4 hours?

2. Part II: Solutions

2.1. Section A Solutions: Advanced Rate Problems

2.1.1. Solution A1: Multi-Leg Journey with Variable Speeds

Let the total distance be D km.

Question 1: Time for each segment in terms of D:
- First third at 60 km/h: $t_1 = \frac{D/3}{60} = \frac{D}{180}$ hours
- Second third at 80 km/h: $t_2 = \frac{D/3}{80} = \frac{D}{240}$ hours
- Final third at 120 km/h: $t_3 = \frac{D/3}{120} = \frac{D}{360}$ hours

Question 2: Total time: $T = \frac{D}{180} + \frac{D}{240} + \frac{D}{360}$

Finding a common denominator (720): $T = \frac{4D}{720} + \frac{3D}{720} + \frac{2D}{720} = \frac{9D}{720} = \frac{D}{80}$ hours

Question 3: Average speed:

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{D}{D/80} = 80 \ \ \text{km/h}$$

The arithmetic mean of 60, 80, and 120 is $\frac{60+80+120}{3} \approx 86.67$ km/h.

The average speed (80 km/h) is less than the arithmetic mean because the car spends **more time** at slower speeds and **less time** at faster speeds. Since each segment covers equal distance, slower segments take longer. The average speed is actually the **harmonic mean** of the three speeds.

Question 4: For $D = 360$ km:
- At variable speeds: time = $\frac{360}{80} = 4.5$ hours
- At constant 86.67 km/h: time = $\frac{360}{86.67} \approx 4.15$ hours
- Difference: $4.5 - 4.15 = 0.35$ hours $\approx$ 21 minutes

Answer: The car would arrive 21 minutes **earlier** at constant speed equal to the arithmetic mean.

2.1.2. Solution A2: Pursuit and Meeting Problems

Question 1: Position functions (using Station A as origin, t in hours):

- Train 1: $x_1(t) = 80t$ km from Station A
- Train 2: $x_2(t) = 450 - 100t$ km from Station A

Question 2: Meeting time and location: $80t = 450 - 100t$ $180t = 450$ $t = 2.5$ hours

Position: $x = 80 \times 2.5 = 200$ km from Station A.

Answer: The trains meet after 2.5 hours, at a point 200 km from Station A.

Question 3: Bird's total distance:

Key Insight: The bird flies continuously at 150 km/h for the entire time until the trains meet. We don't need to track each back-and-forth leg!

Since the trains meet after 2.5 hours, the bird flies for 2.5 hours at 150 km/h:

$$\text{Distance} = 150 \times 2.5 = 375 \quad \text{km}$$

Answer: The bird travels 375 km.

Question 4: With Train 1 departing 30 minutes early: After time t (measured from Train 1's departure):

- Train 1 position: $80t$
- Train 2 position (starts at $t = 0.5$): $450 - 100(t - 0.5)$ for $t \geq 0.5$

At meeting: $80t = 450 - 100(t - 0.5) = 450 - 100t + 50 = 500 - 100t$ $180t = 500$ $t = \frac{500}{180} = \frac{25}{9} \approx 2.78$ hours

Position: $80 \times \frac{25}{9} = \frac{2000}{9} \approx 222.2$ km from Station A.

Answer: They meet approximately 222.2 km from Station A.

2.1.3. Solution A3: Round-Trip Average Speed

Question 1: Proof using harmonic mean:

Let the one-way distance be d km.

- Time uphill: $\frac{d}{12}$ hours
- Time downhill: $\frac{d}{36}$ hours
- Total distance: $2d$ km
- Total time: $\frac{d}{12} + \frac{d}{36} = \frac{3d+d}{36} = \frac{4d}{36} = \frac{d}{9}$ hours

$$\text{Average speed} = \frac{2d}{d/9} = 18 \ \ \text{km/h}$$

This is independent of d, confirming the average is always 18 km/h.

Question 2: General formula:

$$\text{Average speed} = \frac{2d}{\frac{d}{u} + \frac{d}{d}} = \frac{2d}{\frac{d(u+d)}{ud}} = \frac{2ud}{u+d}$$

This is the harmonic mean of u and d.

Question 3: Comparing routes:

Let the hilly route one-way distance = d. Then the flat route distance = $\frac{d}{0.75} = \frac{4d}{3}$.

Hilly round trip time: $\frac{2d}{18} = \frac{d}{9}$ hours

Flat round trip time: $\frac{2 \times \frac{4d}{3}}{20} = \frac{\frac{8d}{3}}{20} = \frac{8d}{60} = \frac{2d}{15}$ hours

Comparing: $\frac{d}{9}$ vs $\frac{2d}{15} = \frac{5d}{45}$ vs $\frac{6d}{45}$

Since $\frac{5d}{45} < \frac{6d}{45}$, the hilly route is faster.

Answer: The hilly route is faster (by $\frac{d}{45}$ hours).

Question 4: For equal times: $\frac{2d}{18} = \frac{2 \times kd}{20}$ where flat = $k \times$ hilly distance $\frac{d}{9} = \frac{kd}{10}$ $10 = 9k$ $k = \frac{10}{9} \approx 1.111$

The flat route would need to be $\frac{10}{9} - 1 = \frac{1}{9} \approx 11.1\%$ longer for equal times.

Answer: The routes take equal time when the flat route is about 11.1% longer.

2.1.4. Solution A4: Stream and Current Problems

Question 1: Setting up the equation:

Downstream speed = $b + c$, upstream speed = $b - c$. Same time for different distances:

$$\frac{30}{b+c} = \frac{18}{b-c}$$

Question 2: Solving for ratio: $30(b - c) = 18(b + c)$ $30b - 30c = 18b + 18c$ $12b = 48c$ $\frac{b}{c} = 4$

Answer: $b : c = 4 : 1$

Question 3: Finding exact values: Downstream: $\frac{48}{b+c} = 2$ hours, so $b + c = 24$ km/h. With $b = 4c$: $4c + c = 24$, giving $c = 4.8$ km/h and $b = 19.2$ km/h.

Answer: Boat speed = 19.2 km/h, current speed = 4.8 km/h.

Question 4: Round trip of 80 km each way:

- Upstream (80 km): $\frac{80}{19.2-4.8} = \frac{80}{14.4} = \frac{50}{9}$ hours
- Downstream (80 km): $\frac{80}{19.2+4.8} = \frac{80}{24} = \frac{10}{3}$ hours
- Total: $\frac{50}{9} + \frac{10}{3} = \frac{50}{9} + \frac{30}{9} = \frac{80}{9} \approx 8.89$ hours

Answer: The round trip takes $\frac{80}{9} \approx 8.89$ hours (8 hours 53 minutes).

2.1.5. Solution A5: Circular Track Problems

Question 1: Meeting at the starting point:

Runner A completes one lap in $\frac{400}{5} = 80$ seconds. Runner B completes one lap in $\frac{400}{3} \approx 133.33$ seconds.

LCM of 80 and $\frac{400}{3}$: Since $\frac{400}{3} = 133.\overline{3}$, we compute LCM$\left(80, \frac{400}{3}\right)$.

Converting: LCM$\left(\frac{240}{3}, \frac{400}{3}\right) =$ LCM$\frac{240,400}{3} = \frac{1200}{3} = 400$ seconds.

Answer: They first meet at the starting point after 400 seconds (6 min 40 sec).

Question 2: First meeting anywhere:

Since they run in opposite directions, their relative speed is $5 + 3 = 8$ m/s. Time to cover one lap at relative speed: $\frac{400}{8} = 50$ seconds.

Answer: They first meet after 50 seconds.

Question 3: Meetings in 10 minutes:

10 minutes = 600 seconds. They meet every 50 seconds. Number of meetings: $\frac{600}{50} = 12$ meetings.

Answer: They meet 12 times in the first 10 minutes.

Question 4: Runner A stops after 4 minutes (240 seconds):

A's position: $5 \times 240 = 1200$ m = $1200 \bmod 400 = 0$ m (back at start!).

Actually, $\frac{1200}{400} = 3$ complete laps, so A is at the starting point.

B's position after 240s: $3 \times 240 = 720$ m counterclockwise = $720 \bmod 400 = 320$ m counterclockwise from start = 80 m clockwise from start.

B needs to travel: $400 - 80 = 320$ m more counterclockwise to reach start. Time: $\frac{320}{3} \approx 106.67$ seconds.

Answer: A is at the starting point. B reaches A approximately 107 seconds after A stops.

2.2. Section B Solutions: Advanced Work Problems

2.2.1. Solution B1: Multiple Workers with Variable Rates

Question 1: Daily rates:

- A: $\frac{1}{10}$ of job per day
- B: $\frac{1}{15}$ of job per day
- C: $\frac{1}{30}$ of job per day

Question 2: All three together: Combined rate = $\frac{1}{10} + \frac{1}{15} + \frac{1}{30}$

LCM of 10, 15, 30 is 30: $= \frac{3}{30} + \frac{2}{30} + \frac{1}{30} = \frac{6}{30} = \frac{1}{5}$ of job per day

Time = 5 days.

Answer: Working together, the job takes 5 days.

Question 3: A and B work 3 days, then B and C finish:

Work done by A and B in 3 days: $3\left(\frac{1}{10} + \frac{1}{15}\right) = 3\left(\frac{3}{30} + \frac{2}{30}\right) = 3\left(\frac{5}{30}\right) = \frac{15}{30} = \frac{1}{2}$

Remaining work: $\frac{1}{2}$

B and C's combined rate: $\frac{1}{15} + \frac{1}{30} = \frac{2}{30} + \frac{1}{30} = \frac{3}{30} = \frac{1}{10}$ per day

Time for remaining: $\frac{1/2}{1/10} = 5$ days

Total: $3 + 5 = 8$ days.

Answer: The job takes 8 days total.

Question 4: Each works exactly 5 days:

Total work done: $5 \times \frac{1}{10} + 5 \times \frac{1}{15} + 5 \times \frac{1}{30} = 5\left(\frac{1}{10} + \frac{1}{15} + \frac{1}{30}\right) = 5 \times \frac{1}{5} = 1$

Yes, exactly 1 complete job!

Answer: Yes, it is possible. If each worker works exactly 5 days (in any arrangement that totals 15 work-days), the job will be completed exactly.

2.2.2. Solution B2: Work with Efficiency Changes

Question 1: Effective hourly rates:

Machine A works 45 min/hour (15 min maintenance), so: $\text{Effective rate}_A = 120 \times \left(\frac{45}{60}\right) = 120 \times 0.75 = 90$ widgets/hour

Machine B works continuously: 80 widgets/hour.

Question 2: 8-hour shift production:

Machine A: $8 \times 90 = 720$ widgets Machine B: $8 \times 80 = 640$ widgets Total: $720 + 640 = 1,360$ widgets

Answer: 1,360 widgets are produced in an 8-hour shift.

Question 3: Machine B with efficiency drop:

First 4 hours: $4 \times 80 = 320$ widgets Next 4 hours at 75% efficiency: $4 \times 60 = 240$ widgets Total for B: $320 + 240 = 560$ widgets

Answer: Machine B produces 560 widgets in the 8-hour shift.

Question 4: Time for 1,400 widgets:

For the first 4 hours:
- Combined rate: $90 + 80 = 170$ widgets/hour
- Production: $4 \times 170 = 680$ widgets
- Remaining: $1400 - 680 = 720$ widgets

After 4 hours:
- Combined rate: $90 + 60 = 150$ widgets/hour
- Time for 720 widgets: $\frac{720}{150} = 4.8$ hours

Total time: $4 + 4.8 = 8.8$ hours

Answer: The order is complete after 8.8 hours (8 hours 48 minutes).

2.2.3. Solution B3: Pipes and Tanks

Question 1: Hourly rates:

- Pipe A: $+\frac{1}{4}$ tank/hour (fills)
- Pipe B: $+\frac{1}{6}$ tank/hour (fills)
- Drain: $-\frac{1}{8}$ tank/hour (empties)

Question 2: All three open, starting empty:

Net rate = $\frac{1}{4} + \frac{1}{6} - \frac{1}{8}$

LCM of 4, 6, 8 is 24: $= \frac{6}{24} + \frac{4}{24} - \frac{3}{24} = \frac{7}{24}$ tank/hour

Time to fill: $\frac{1}{7/24} = \frac{24}{7} \approx 3.43$ hours

Answer: The tank fills in $\frac{24}{7} \approx 3.43$ hours (about 3 hours 26 minutes).

Question 3: Tank starts $\frac{1}{3}$ full; A and drain open for 2 hours, then B joins:

First 2 hours (A and drain only): Rate = $\frac{1}{4} - \frac{1}{8} = \frac{2}{8} - \frac{1}{8} = \frac{1}{8}$ tank/hour Added in 2 hours: $2 \times \frac{1}{8} = \frac{1}{4}$ tank Tank level after 2 hours: $\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$

Remaining to fill: $1 - \frac{7}{12} = \frac{5}{12}$

After B joins (all three open): Rate = $\frac{7}{24}$ tank/hour (from Question 2) Time: $\frac{5/12}{7/24} = \left(\frac{5}{12}\right) \times \left(\frac{24}{7}\right) = \frac{10}{7}$ hours

Total time: $2 + \frac{10}{7} = \frac{24}{7} \approx 3.43$ hours

Answer: The tank is full after $\frac{24}{7} \approx 3.43$ hours from the start.

Question 4: Close drain at time t_0 so tank fills at exactly $t = 6$ hours:

For $0 \leq t \leq t_0$: rate = $\frac{7}{24}$, fill = $\frac{7t_0}{24}$ For $t_0 < t \leq 6$: rate = $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$, fill = $\left(\frac{5}{12}\right)(6 - t_0)$

Total: $\frac{7t_0}{24} + \left(\frac{5}{12}\right)(6 - t_0) = 1$

$\frac{7t_0}{24} + \frac{30-5t_0}{12} = 1$ $\frac{7t_0}{24} + \frac{60-10t_0}{24} = 1$ $\frac{7t_0+60-10t_0}{24} = 1$ $\frac{60-3t_0}{24} = 1$ $60 - 3t_0 = 24$ $t_0 = 12$ hours

But this exceeds 6 hours! This means the tank fills before 6 hours even with the drain always open. The drain should remain open the entire time; closing it earlier would only speed up filling.

Answer: The tank fills in $\frac{24}{7} < 6$ hours with drain always open, so closing the drain is not necessary (or possible within the 6-hour window).

2.3. Section C Solutions: Mixture Problems

2.3.1. Solution C1: Multi-Component Solutions

Question 1: A and B for 35% solution:

Let x mL of A and $(100 - x)$ mL of B. $0.15x + 0.40(100 - x) = 0.35(100)$ $0.15x + 40 - 0.40x = 35$ $-0.25x = -5$ $x = 20$ mL of A, 80 mL of B

Answer: 20 mL of Solution A and 80 mL of Solution B.

Question 2: Equal volumes of all three:

Each = $\frac{100}{3}$ mL (total 100 mL) Concentration = $\frac{0.15+0.40+0.75}{3} = \frac{1.30}{3} \approx 0.433 = 43.3\%$

This is greater than 35%, so equal volumes cannot achieve 35%.

Answer: No. Equal volumes produce 43.3% concentration, not 35%.

Question 3: 300 mL at 50% with volume of A = volume of C:

Let A = C = x mL, B = $(300 - 2x)$ mL. $0.15x + 0.40(300 - 2x) + 0.75x = 0.50(300)$ $0.15x + 120 - 0.80x + 0.75x = 150$ $0.10x + 120 = 150$ $0.10x = 30$ $x = 300$

But this gives B = $300 - 600 = -300$, which is impossible!

The constraint that A = C is incompatible with achieving 50%. We need to reconsider.

Average of A and C: $\frac{15+75}{2} = 45\%$. Adding B (40%) can only lower this toward 40%, never reach 50%.

Answer: This is impossible. Equal volumes of A and C average 45%, and adding B (40%) cannot raise the concentration to 50%.

Question 4: Range using A and C only:

Minimum: 15% (pure A) Maximum: 75% (pure C) All values in between are achievable by linear combination.

Answer: Any concentration in the interval [15%, 75%] is achievable.

2.3.2. Solution C2: Repeated Replacement

Question 1: First replacement:

Initial: 200 L at 60% = 120 L alcohol Remove 40 L of 60% solution: remove $40 \times 0.60 = 24$ L alcohol Remaining alcohol: $120 - 24 = 96$ L Add 40 L water: total volume returns to 200 L New concentration: $\frac{96}{200} = 48\%$

Answer: The concentration is 48% after the first replacement.

Question 2: Second replacement:

After first: 200 L at 48% = 96 L alcohol Remove 40 L of 48% solution: remove $40 \times 0.48 = 19.2$ L alcohol Remaining: $96 - 19.2 = 76.8$ L alcohol New concentration: $\frac{76.8}{200} = 38.4\%$

Answer: The concentration is 38.4% after the second replacement.

Question 3: Formula after n operations:

Each operation retains $\frac{200-40}{200} = \frac{160}{200} = 0.8$ of the alcohol. After n operations: concentration = $60\% \times (0.8)^n$

Answer: $C_n = 60 \times (0.8)^n$ percent

Question 4: Operations to reach below 20%:

$60 \times (0.8)^n < 20$ $(0.8)^n < \frac{1}{3}$

Taking logarithms: $n > \frac{\log(1/3)}{\log(0.8)} = \frac{\log(3)}{\log(1.25)} \approx \frac{1.099}{0.223} \approx 4.9$

So $n \geq 5$ operations.

Check: $60 \times (0.8)^5 = 60 \times 0.32768 \approx 19.66\% < 20\%$ ✓

Answer: 5 operations are needed.

2.4. Section D Solutions: Age Problems

2.4.1. Solution D1: Multiple Time References

Question 1: System of equations:

Let M = Maria's current age, E = Elena's current age.

- Currently: $M = 4E$
- In 16 years: $M + 16 = 2(E + 16)$

Question 2: Solving:

From equation 2: $M + 16 = 2E + 32$, so $M = 2E + 16$ Substituting into equation 1: $4E = 2E + 16$ $2E = 16$ $E = 8$, $M = 32$

Answer: Maria is 32 years old; Elena is 8 years old.

Question 3: When was Maria 7 times Elena's age?

$32 - x = 7(8 - x)$ $32 - x = 56 - 7x$ $6x = 24$ $x = 4$ years ago

Answer: 4 years ago (when Maria was 28 and Elena was 4).

Question 4: When will sum exceed 100?

$(32 + y) + (8 + y) > 100$ $40 + 2y > 100$ $y > 30$

The sum first exceeds 100 after 31 years (when sum = 102).

Answer: In 31 years, the sum of their ages will first exceed 100.

2.4.2. Solution D2: Age Ratios

Question 1: Expressing ages:

Let ages be k, $2k$, and $4k$ (ratio 1:2:4).

Question 2: Solving:

$k^2 + (2k)^2 + (4k)^2 = 336$ $k^2 + 4k^2 + 16k^2 = 336$ $21k^2 = 336$ $k^2 = 16$ $k = 4$

Ages: 4, 8, 16 years.

Answer: The siblings are 4, 8, and 16 years old.

Question 3: When will ratio be 2:3:5?

$(4 + y) : (8 + y) : (16 + y) = 2 : 3 : 5$

From first two terms: $\frac{4+y}{2} = \frac{8+y}{3}$ $3(4 + y) = 2(8 + y)$ $12 + 3y = 16 + 2y$ $y = 4$

Check third term: $\frac{16+4}{5} = \frac{20}{5} = 4$ ✓ And $\frac{8+4}{3} = \frac{12}{3} = 4$ ✓

Answer: In 4 years, their ages (8, 12, 20) will be in ratio 2:3:5.

Question 4: Can ratio be 3:4:5?

$(4 + y) : (8 + y) : (16 + y) = 3 : 4 : 5$

From first two: $\frac{4+y}{3} = \frac{8+y}{4}$ $4(4 + y) = 3(8 + y)$ $16 + 4y = 24 + 3y$ $y = 8$

Check: Ages would be 12, 16, 24. Ratio: 12:16:24 = 3:4:6 (not 3:4:5!)

The third term doesn't match because $\frac{24}{5} = 4.8 \leq 4$.

Answer: No. The ratio 3:4:5 is impossible (when the first two match 3:4, the third gives 3:4:6).

2.4.3. Solution D3: Past and Future Conditions

Question 1: Setting up equations:

Let F = father's current age, S = son's current age.

- Five years ago: $F - 5 = 7(S - 5)$
- Five years from now: $F + 5 = 3(S + 5)$

Question 2: Solving:

From equation 1: $F = 7S - 35 + 5 = 7S - 30$ From equation 2: $F = 3S + 15 - 5 = 3S + 10$

Setting equal: $7S - 30 = 3S + 10$ $4S = 40$ $S = 10$, $F = 40$

Answer: The father is 40 years old; the son is 10 years old.

Question 3: When was father 5 times son's age?

$40 - x = 5(10 - x)$ $40 - x = 50 - 5x$ $4x = 10$ $x = 2.5$ years ago

Father was 37.5, son was 7.5.

Answer: 2.5 years ago, when the father was 37.5 and son was 7.5 years old.

Question 4: Could father ever be 10 times son's age?

$40 - x = 10(10 - x)$ $40 - x = 100 - 10x$ $9x = 60$ $x = \frac{20}{3} \approx 6.67$ years ago

Check: Son's age then: $10 - \frac{20}{3} = \frac{10}{3} \approx 3.33$ years old (positive, so valid!) Father's age: $40 - \frac{20}{3} = \frac{100}{3} \approx 33.33$ years old

Verification: $33.33 = 10 \times 3.33$ ✓

The statement in the problem is incorrect! The father WAS 10 times the son's age.

Answer: Actually, the father WAS 10 times the son's age, approximately 6.67 years ago (when son was 3.33 and father was 33.33 years old).

2.4.4. Solution D4: Combined Age Conditions

Question 1: Three equations:

Let G = grandfather, F = father, S = son.

- Sum: $G + F + S = 120$
- Ratio: $G = 2F$
- Past condition: $G - 10 = 3(F - 10)$

Question 2: Solving:

From equation 3: $G - 10 = 3F - 30$, so $G = 3F - 20$ From equation 2: $G = 2F$

Setting equal: $2F = 3F - 20$ $F = 20$

Wait, this gives $G = 40$, but then $G - 10 = 30$ and $3(F - 10) = 3(10) = 30$ ✓

From equation 1: $40 + 20 + S = 120$, so $S = 60$

But $S = 60 > F = 20$? A son older than father is impossible!

Let me recheck… The problem may have an inconsistency, or we interpret it differently.

Actually, with $G = 40$, $F = 20$, a 20-year-old father with a 60-year-old son is indeed impossible.

The problem as stated has no realistic solution.

Answer: The given conditions are inconsistent (they imply $G = 40$, $F = 20$, $S = 60$, where the son is older than the father). The problem has no valid real-world solution.

Questions 3 and 4: Cannot be answered due to inconsistent premises.